

Class	T.Y. B. Tech Sem VI
Course Code & Course Title	2CSPC313 Machine Learning
Prerequisite/s	2CSPC201 Discrete mathematics 2CSVS308 Python Programming
Teaching Scheme (Lecture/Practical/Tutorial)	03/00/02
Credits	04
Evaluation Scheme Theory : ISE/MSE/ESE:	40/30/30
Evaluation Scheme Practical: ISE/ESE	50/50

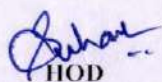
Course Outcomes (COs) :

Upon successful completion of this course, the student will be able to:

2CSPC313_1	Apply various machine learning algorithms to real-world datasets for solving classification, regression, and clustering problems.
2CSPC313_2	Demonstrate the working of various machine learning algorithms using mathematical justifications
2CSPC313_3	Analyze the strengths and weaknesses of different machine learning algorithms for specific types of problems and datasets.
2CSPC313_4	Evaluate machine learning model using appropriate metrics and perform hyper parameter tuning to improve performance.

Course Contents:

Unit 1	Introduction Introduction to Machine Learning, Applications, History of machine learning, Types of Learning, Hypothesis space, Inductive Bias, Data Partitioning Methods, Performance evaluation	06 Hrs.
Unit 2	Regression and Decision Trees Types of Regression - Simple, Multiple, Linear, Non-linear, Gradient Descent and Normal Equations, Polynomial Regression, Logistic Regression, Regularization. Decision Tree representation - ID3 algorithm, Issues	07 Hrs.
Unit 3	Instance Based Learning and Feature Selection k-nearest neighbor, Distance weighted nearest neighbor algorithm, Curse of Dimensionality, Feature selection - filter methods, wrapper methods, Feature extraction - PCA	07 Hrs.
Unit 4	Probability and Bayes Learning Probability Concepts, Bayes Theorem, MAP Hypothesis, Bayes Optimal Classifier, Naive Bayes Classifier, Bayesian Network	05 Hrs.


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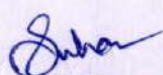


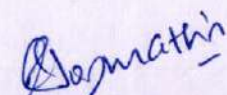
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Unit 5	Support Vector Machines and Neural Network Introduction to support Vector Machine, Linear SVM, Non-linear SVM, Kernel Functions, Multi-class SVMs Introduction to neural network, Perceptron, Perceptron Learning, Multilayer neural network, back propagation, Introduction to deep neural network	08 Hrs.
Unit 6	Clustering, Recommender Systems and Ensemble Learning Introduction to clustering, k-means clustering, Hierarchical Clustering, Recommender Systems - Content based, collaborative filtering, Introduction to ensemble learning, Bagging and boosting, Random Forest Algorithm	06 Hrs.

List of Experiments

1	Program based on Numpy and pandas
2	Implementation of simple linear regression using scikit-learn.
3	Implementation of Logistic Regression for classification
4	Build the Decision Tree Model for given problem statement (Use ID3 Algorithm) - Use Pen and Paper
5	Implementation of decision tree classifier in python. Use of scikit-learn for various functionalities.
6	Problem solving - Naive Bayes classifier.
7	Implementation of Naive Bayes classifier in python. Use of scikit-learn for various functionalities.
8	Implementation of &-NN algorithm in python. Use of scikit-learn for various functionalities.
9	Implementation of ^-means Clustering.
10	Implementation of SVM for classification.
11	Program based on neural network


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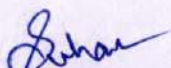
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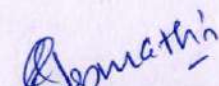
Text Books:

Sr. No	Title	Author	Publisher	Edition	Year of Edition
01	Machine Learning	Tom Mitchell	McGraw-Hill	1	1997
02	Introduction to Machine Learning	Ethem Alpaydin	The MIT Press	2	2010

Reference Books:

Sr. No	Title	Author	Publisher	Edition	Year of Edition
01	The Elements of Statistical Learning	T. Hastie, R. Tibshirani, J. Friedman	Pearson Education	2	2008
02	NPTEL course Introduction to Machine Learning	Prof. Sudeshna Sarkar	NPTEL	-	-
03	Coursera Machine Learning	Andrew Ng	Coursera / Stanford University	-	-


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